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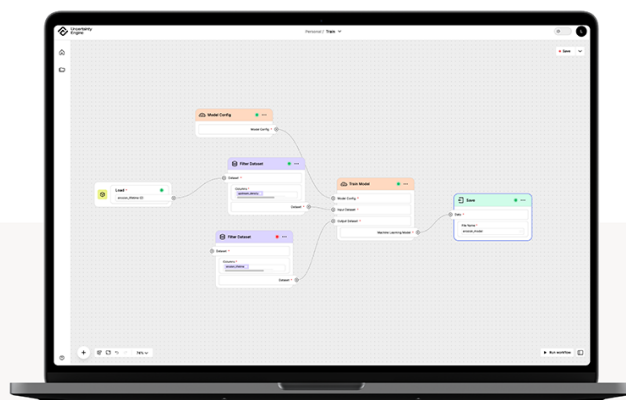


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Unified ELM suppression on KSTAR and DIII-D via adaptive feedback control strategies

R. Shousha^{1,7} , S.K. Kim^{1,7} , K.G. Erickson¹, S.H. Hahn² , A.O. Nelson³ ,
N.C. Logan³ , S.M. Yang¹ , Q.M. Hu¹ , R. Wilcox⁴ , J.-K. Park¹ , C. Paz-Soldan³ ,
A. Battey⁵ , M.W. Kim² , G. Shin² , W.H. Ko² , Y.M. Jeon² , J.H. Lee² , J. Jang² ,
D.C. Seo², J. Abbate¹ , A. Rothstein⁶  and E. Kolemen^{1,6,*} 

¹ Princeton Plasma Physics Laboratory, Princeton, NJ 08543, United States of America

² Korea Institute of Fusion Energy, Daejeon 34133, Korea, Republic Of

³ Columbia University, New York, NY 10027, United States of America

⁴ Oak Ridge National Laboratory, Oak Ridge, TN 37830, United States of America

⁵ Swiss Plasma Center, École Polytechnique Fédérale de Lausanne (EPFL), CH-1015 Lausanne, Switzerland

⁶ Princeton University, Princeton, NJ 08544, United States of America

E-mail: ekolemen@pppl.gov and rshousha@pppl.gov

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Abstract

This paper reports on the extension of our amplitude-based resonant magnetic perturbation (RMP) edge localized mode (ELM) controller to support phasing control (relative toroidal phases of RMP waveforms between rows of coils), multiple toroidal mode numbers, and new ‘jump’ and ‘probing’ strategies, all deployed on KSTAR and DIII-D. By treating the control algorithm as device-independent and adjusting only the real-time interfaces to sensors and power supplies, we have confirmed that the same finite state machine—based feedback logic can be ported between machines with minor modifications. In experiments using $n = 2$ RMPs on KSTAR and $n = 3$ on DIII-D, the controller successfully modulated RMP amplitudes in real time to sustain ELM suppression while minimizing confinement degradation. Phasing control broadened the suppression window, as it permitted the system to avoid locked-mode regions and safely access ELM-free conditions. A rotating RMP phasing scheme, integrated into the same framework, distributes divertor heat loads more uniformly, making it a promising strategy for protecting plasma-facing components during long discharges. New ‘jump’ and ‘probing’ techniques demonstrate the possibility for the controller to preempt imminent ELMs and refine the minimum required RMP amplitude without returning to ELM conditions. Taken together, these upgrades enable extended ELM-free operation while mitigating confinement degradation, providing a practical framework for real-time ELM control in future high-performance tokamaks.

⁷ These authors contributed equally.

* Author to whom any correspondence should be addressed.



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Keywords: ELM suppression, RMP, adaptive feedback control, fusion plasma, KSTAR, DIII-D, feedback

(Some figures may appear in colour only in the online journal)

1. Introduction

One of the promising paths to economically viable tokamak power plants is to operate the plasma in the H-mode regime [1]. The edge transport barrier characteristic of this mode elevates the pressure profile for improved fusion conditions and forms a pedestal structure in the plasma edge [2]. These favorable confinement conditions are limited by the destabilization of edge localized modes (ELMs)—a quasi-periodic expulsion of heat and particles to the first wall and divertor [3, 4]. Unfortunately, when tokamaks are scaled up to the size of the ITER device (which also relies on H-mode [5, 6]) or beyond, the heat load associated with a single ELM will exceed current material limits [5, 7, 8]. This necessitates the development of ELM control techniques that can mitigate the deleterious effects of ELMs while minimizing any impact on fusion performance in high-power H-mode plasmas.

The application of specific non-axisymmetric edge resonant three-dimensional (3D) magnetic field perturbations (RMPs) using toroidally distributed in-vessel coils has been widely observed to suppress ELMs [9–14]. This suppression is achieved by enabling additional transport in the pedestal [15–28], which limits the pedestal height below the ELM stability limit [29–31]. In addition, earlier experiments demonstrated that a hysteresis phenomenon can be exploited to achieve optimized ELM suppression [32–34]: The RMP amplitude required to *enter* ELM suppression is greater than that required to *sustain* it, and performance parameters like confinement are inversely proportional to RMP amplitude. Therefore, the use of RMP actuators is a promising candidate for active ELM control.

However, there are no real-time physics-based models available yet that can predict the minimum RMP amplitude required to sustain ELM suppression. To address this problem, the ongoing development of a feedback adaptive RMP ELM controller is reported in this work. The controller uses real-time D_α measurements to monitor ELM activity. Then, using a finite state machine design, the controller ‘ramps up’ (linearly increases), holds steady, or ‘ramps down’ the quantity controlled in feedback. This mechanism is generic and can be applied to both amplitude and phasing feedback. Note that when we refer to phasing, we are talking about the difference in toroidal phase between rows of RMP coil sets. This concept is mathematically defined and visually explained in [33]. The controller computes the required coil current amplitudes and phases, and translates these into power supply commands to control the local controllers of the 3D RMP coils of the tokamak. The main design of the controller and more details about results obtained in previous KSTAR experiments can be found in earlier published work by Shousha *et al* [33] and Kim *et al* [34]. The KSTAR experiments demonstrated

real-time controlled ELM suppression with minimization of confinement degradation. However, inherent to the design, the existing controller has to *lose* ELM suppression at least once to optimize the RMP amplitude and find the lower bound. Additionally, in long pulses where the discharge parameters evolve significantly, the lower bound that the controller computes might become outdated and require an update. Furthermore, to increase confidence in ITER-scale applicability, exploration of control at higher toroidal mode numbers than $n = 1$ RMP and of cross-device applicability is needed.

Therefore, this paper reports on control improvements in several key areas: generalizing the controller for increased flexibility and robustness, expanding the ELM suppression window in feedback control, achieving complete ELM-suppressed H-mode discharges, integrating control strategies for long-pulse ELM suppression, and addressing remaining challenges in ELM suppression. We have extended the application domain by porting the feedback adaptive RMP ELM controller from KSTAR to the DIII-D plasma control system (PCS) with minimal differences, verifying cross-device performance. Control capabilities now include higher toroidal mode numbers ($n = 2$ and $n = 3$), phasing control beyond amplitude adjustment, and the application of rotating RMP fields, which may benefit toroidal heat load distribution in long pulses. To enhance the ELM suppression window in feedback control, we explore the history effect through phasing feedback. To achieve complete ELM-free H-mode discharges, we have developed a new feedback initialization capability and a real-time precursor detector based on an Interactive Multiple Model Kalman filter approach [35, 36], enabling the controller to act proactively before an imminent ELM. Integrating these control strategies supports long-pulse ELM suppression and mitigates RMP-induced asymmetric heat flux through rotating fields and adaptive lower-bound reduction. Finally, we address remaining challenges in ELM suppression, discussing ongoing issues and potential future research directions.

The results presented here underscore the benefits of phasing feedback control. Building on these advances, we have recently demonstrated that the same control framework can be supplemented with a machine—learning—based RT-GPEC model, which provides phasing and amplitude ratio commands for simultaneous amplitude and phasing feedback. This approach achieves and maintains ELM suppression with minimal impact on confinement and enables high fusion performance during ELM-free conditions, as detailed in [37].

This paper is structured as follows. Section 2 focuses on the generalization of control strategies across fusion devices, including adaptive amplitude control and phasing control. Section 3 discusses expanding the ELM suppression window in feedback control. Section 4 details efforts toward complete ELM suppression, including early ELM suppression enhanced

by machine learning (ML), precursor detection techniques, the jump-scheme, and the probing-scheme. Section 5 covers integrating control strategies toward long-pulse ELM suppression, including phasing control to mitigate RMP-induced asymmetric heat flux and adaptive lower-bound reduction. Section 6 addresses remaining challenges in ELM suppression. Conclusions are presented in section 7.

2. Generalization of control strategies across fusion devices

Developing a robust and flexible ELM control strategy requires a generic and modular solution that can be deployed across different tokamak devices without significant redesign. The control system is divided into three main components: sensors, the control algorithm, and actuators. By maintaining a single control algorithm and adapting only the sensors and actuators to the specific device, we can generalize the controller effectively.

The primary real-time diagnostic used by the controller is a photodiode measuring the D_α radiation, which indicates ELM activity. Both KSTAR and DIII-D have such diagnostics available at a 20 kHz sampling rate, but with different lines of sight. On KSTAR, the default sensor is aimed at the upper divertor (PCPOLHA02), while on DIII-D, a top-down line of sight aimed at the lower divertor (FS02DA) is used (see figure 1).

Despite differences in the lines of sight, the global nature of ELMs ensures that clear signals are obtained on both devices. Figure 1 shows the ELM signatures captured by the photodiodes on KSTAR and DIII-D during similar time intervals. The ELM signals are fundamentally similar between the two devices, allowing existing ELM detection tools (e.g. [38]) to be ported without requiring significant redesign. Some parameters may need adjustment to account for signal amplitude differences, but overall, the controller can effectively use photodiode measurements from different devices.

The actuators available for 3D ELM control differ significantly between KSTAR and DIII-D. On KSTAR, the in-vessel control coils consist of three rows of four toroidally distributed coils each, allowing for flexible RMP configurations. In contrast, DIII-D typically uses the I-coils, which are composed of two rows of six coils each, spanning the toroidal circumference (see figure 2).

The differences in coil design and the number of physical coils affect the potential RMP spectra that can be generated. DIII-D's configuration allows for higher toroidal mode numbers (e.g. $n = 3$), while KSTAR's setup is typically used for lower mode numbers. Additionally, the number of available power supplies and how the coils are connected (e.g. in series) further influence the RMP configurations achievable on each device.

Despite these differences, the controller is designed to accommodate various actuator configurations by utilizing a device-specific backend that translates control commands into the appropriate power supply inputs. This modular approach ensures that the same control algorithm can be used across different devices, with adaptations made only to the actuator

interface. By maintaining consistency in the control strategy while adjusting for hardware differences, we enhance the flexibility and robustness of the ELM suppression controller across fusion devices.

2.1. Adaptive amplitude control

A key requirement for robust ELM suppression is maintaining an optimal RMP amplitude in real time. Adjusting the RMP amplitude allows the plasma pedestal to remain ELM-free, but too high an RMP amplitude can degrade confinement, whereas too low can lead to loss of ELM suppression. Extending this balance to various toroidal mode numbers (n) further broadens the operational space and is crucial for future large-scale devices like ITER, where $n = 3$ fields are anticipated [39]. The controller described here achieves adaptive amplitude control by continuously sensing the plasma state via D_α emission and updating the RMP current requests in ways that sustain ELM-free H-mode while preserving as much confinement as possible.

In practice, the choice of n depends on coil design and machine capabilities. KSTAR's coil geometry supports $n = 1$ and $n = 2$ fields, whereas DIII-D can access $n = 1$, $n = 2$, and $n = 3$. To address the rising importance of $n > 1$ modes for next-generation devices, the adaptive controller was extended from its initial $n = 1$ implementation [33] to higher n -numbers. This section demonstrates its $n = 2$ capability on KSTAR and its $n = 3$ functionality on DIII-D. By enabling feedback-based ELM suppression across multiple n -values, we can systematically explore the effect of RMP geometry on pedestal stability, density pump-out, and overall plasma performance.

2.1.1. Demonstration of $n = 2$ feedback controlled ELM suppression on KSTAR. On KSTAR, we first assessed the ELM suppression window for an $n = 2$ configuration using a feed-forward ramp of the RMP amplitude (i.e. we manually scanned the RMP current to see where ELMs vanished). Major plasma parameters for these experiments were $I_p \approx 0.6$ MA, $B_T \approx 1.8$ T, $q_{95} \approx 3.9$, and $\kappa \approx 1.7$. Once a suitable amplitude range was identified, we enabled the adaptive amplitude control to regulate the RMP current in real time, always seeking the lowest amplitude that still maintained ELM suppression.

Figure 3 shows a representative discharge using $n = 2$ feedback. Panel (a) displays the D_α signal in black and shades intervals of detected ELM suppression in orange. The controller monitors the plasma state continuously and updates its internal states (RAMPUP, FLAT, and RAMPDOWN—labeled RU, FLT, and RD respectively), which are plotted in blue in panel (b). The rapid switching between RAMPDOWN and FLAT occurs because we instructed the controller to spend minimal time in the FLAT state, after it completes a pre-defined reduction in RMP amplitude in the RAMPDOWN state (it will only monitor ELM suppression outside of the RAMPDOWN state, and in this experiment, we did not want the controller to spend much time in the FLAT state between RAMPDOWN phases). The requested

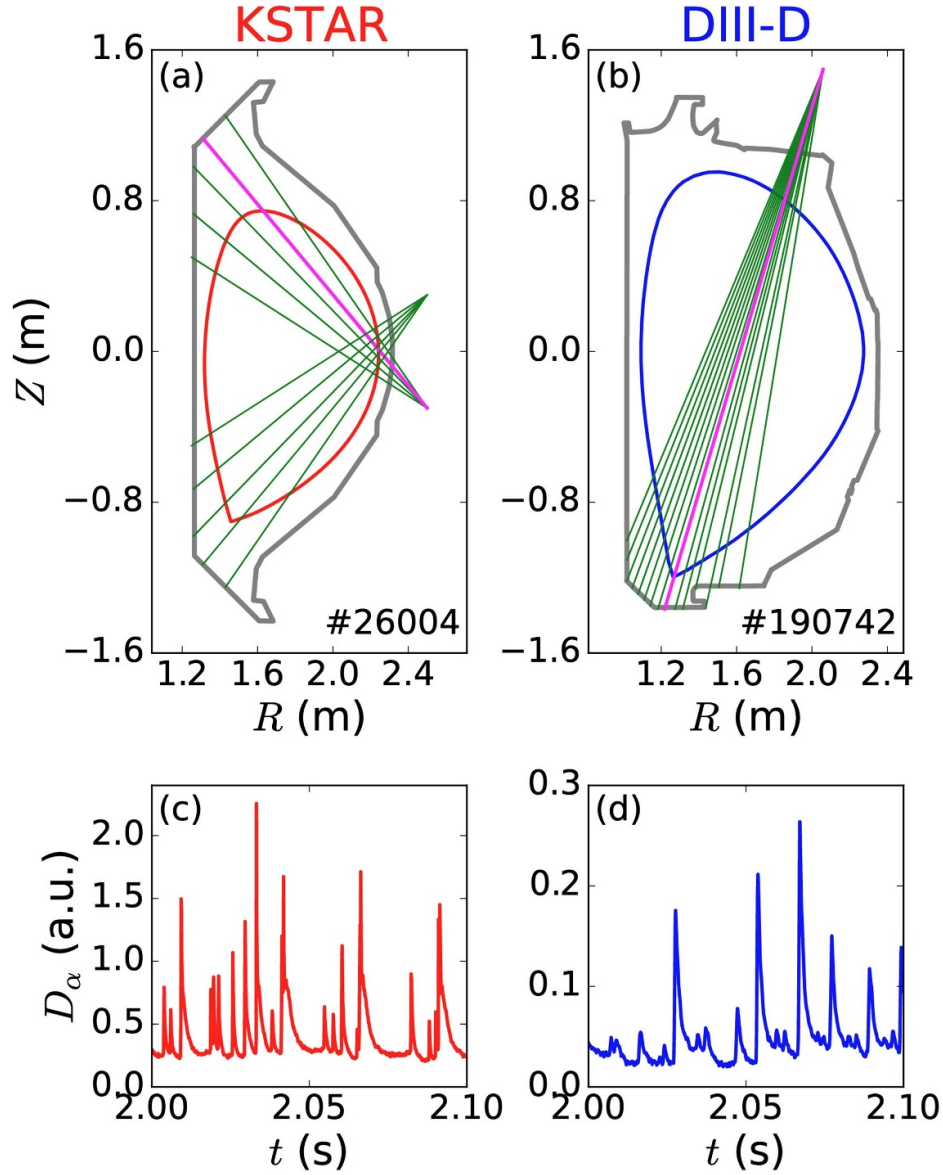


Figure 1. Schematic representation of lines of sight of photo diodes on the KSTAR and DIII-D tokamak devices.

RMP amplitude in red reveals how the controller systematically ramps up or down to balance ELM avoidance with confinement recovery. Panel (c) tracks the pedestal electron density (green) and the normalized plasma pressure β_N (purple).

During the flat phases, the amplitude is held constant to confirm the plasma remains ELM-free. When the controller detects stable conditions over a preset duration, it transitions to RAMPDOWN, lowering the amplitude to reduce RMP-related confinement penalties. If a precursor or ELM crash appears, the state changes to RAMPUP to restore a sufficiently high perturbation. Notably, the controller avoided ELMs for an extended period but eventually observed a loss of suppression at roughly half the initial amplitude. It responded by increasing the amplitude until ELMs were again suppressed. While this $n = 2$ scenario did not fully recover the same high confinement seen in $n = 1$ cases, it confirmed that the adaptive amplitude approach can robustly stabilize ELMs at higher mode

numbers. Ongoing work focuses on refining coil phasing and feedback thresholds to further optimize performance under $n = 2$ conditions.

2.1.2. Demonstration of $n = 3$ feedback controlled ELM suppression on DIII-D. Because KSTAR cannot produce an $n = 3$ field with its available coils, we tested the controller's higher- n capability on DIII-D. In these experiments, the same adaptive amplitude logic was integrated into the DIII-D PCS. Figure 4 illustrates a discharge (with $I_p \approx 1.6$ MA, $B_T \approx -1.9$ T, $q_{95} \approx 3.3$, and $\kappa \approx 1.8$) in which $n = 3$ ELM suppression was successfully achieved under amplitude feedback. As in the KSTAR trials, the RMP amplitude was adjusted dynamically to suppress ELMs while minimizing detrimental confinement degradation or core-locking.

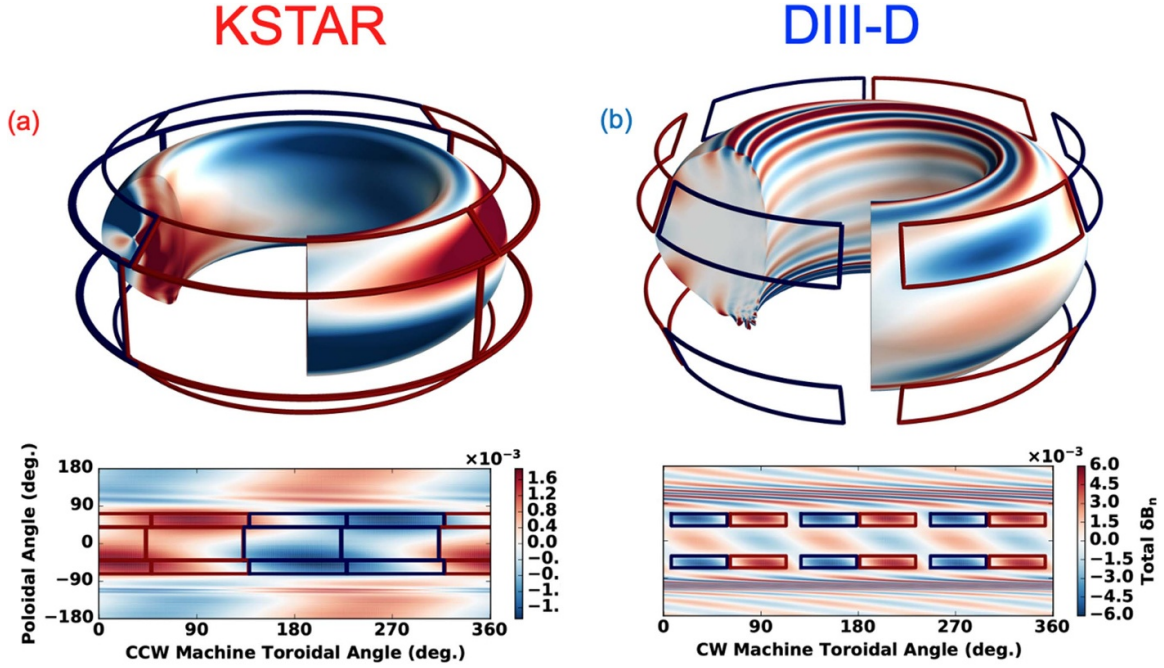


Figure 2. The coils available for RMP ELM control on KSTAR are the so-called in-vessel Correction Coils (IVCCs). As shown in the left panel (a), KSTAR employs three rows of four coils each. In contrast, the 3D coils typically used for RMP ELM control on DIII-D (I-coils) consist of two rows of six coils each, spanning the toroidal circumference.

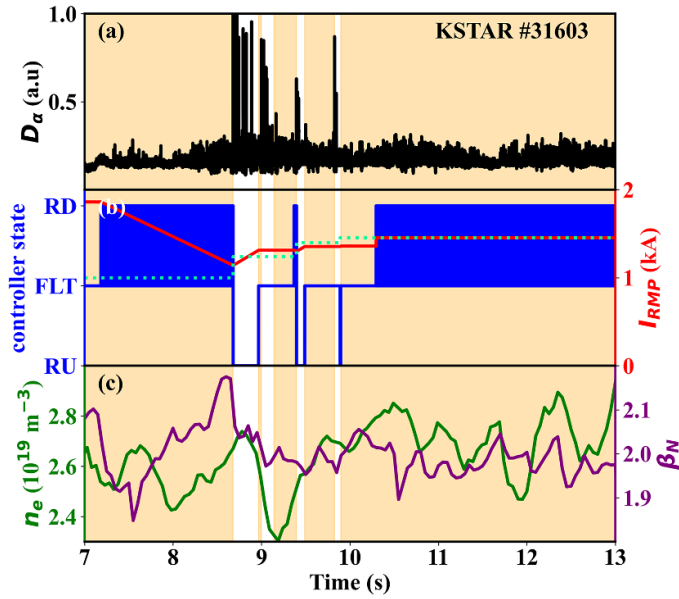


Figure 3. Initial demonstration of $n = 2$ ELM suppression capability of the adaptive RMP controller on KSTAR. (a) The D_α trace is shown in black, with ELM-suppressed intervals shaded in orange. (b) The controller's internal state is plotted in blue, and the RMP amplitude request in red. (c) Pedestal electron density (green) and normalized plasma pressure (purple). Note that the controller is not permitted to ramp down further after 10.5 s, even though the state machine attempts to do so. This is prevented by the adaptive lower bound (shown in bright green, dashed), which is determined through feedback when ELM suppression is lost. This mechanism ensures that the RMP amplitude does not fall below the minimum required for ELM suppression.

Panel (a) shows the D_α signal in black, with orange-shaded intervals marking detected ELM-free periods. Panel (b) presents the requested RMP amplitude (red) alongside the controller's state (blue), revealing short downward RMP

'probes' that let the system test whether reduced amplitude still keeps the pedestal stable. Panel (c) plots the measured electron pedestal density (green) and the normalized plasma pressure (purple). Notably, even with these probes, the

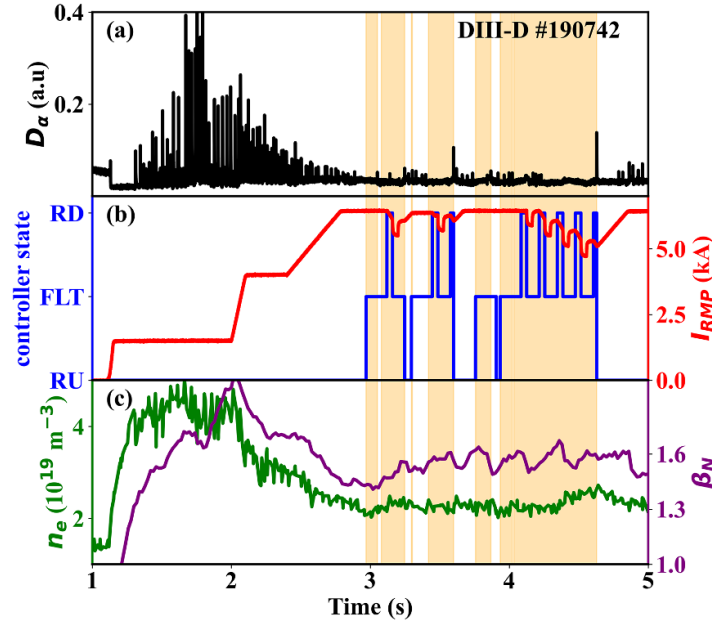


Figure 4. Initial demonstration of $n = 3$ feedback-controlled ELM suppression on DIII-D. (a) The D_α trace (black) shows ELM-suppressed intervals in orange. (b) The RMP amplitude request (red) includes short downward probes, while the controller's state is shown in blue. (c) Pedestal density (green) and β_N (purple).

controller maintained ELM-free conditions most of the time, demonstrating the potential of $n = 3$ amplitude control in a separate tokamak environment.

These results confirm that the same adaptive amplitude framework can handle different machines and different toroidal mode number configurations with only minor modifications. Over multiple discharges, the system's ability to mitigate ELMs in the presence of $n = 3$ fields proved robust, albeit with some trade-offs for core confinement that merit further investigation. Future discharges will incorporate additional features—such as the jump-scheme for immediate RMP ‘pushes’ when precursors arise—to better balance the need for ELM suppression with minimal confinement degradation.

2.1.3. Implications for multi- n operation and future devices.

Taken together, the $n = 2$ experiments on KSTAR and $n = 3$ experiments on DIII-D highlight the versatility of adaptive amplitude control. Since ITER is expected to rely heavily on $n = 3$ RMP coils [39], demonstrating ELM suppression at the same mode number in present-day devices is an important step toward bridging the gap between mid-sized tokamaks and reactor-scale systems. Meanwhile, other facilities may choose $n = 2$ or $n = 1$ configurations, so having a single feedback algorithm that adapts to various n -values ensures broader applicability across the fusion community.

An important future direction involves combining these multi- n amplitude strategies with real-time phasing adjustments, which could help distribute divertor heat loads more evenly and further mitigate localized erosion. Additionally, synergy with ‘probing-schemes’ allows the controller to

gradually relax the lower amplitude bound without triggering full-blown ELMs, boosting overall device performance and pulse length.

Note that in the current experiments, the toroidal mode number n is selected prior to the discharge based on coil geometry and patch panel configuration. Dynamic switching of n during a shot is not yet supported due to hardware limitations, as coils are grouped and powered by shared supplies. However, this constraint is not fundamental; future devices with individually powered coils could allow the control algorithm to dynamically choose the optimal n in real time.

By dynamically matching RMP fields to the plasma's evolving stability margin—across $n = 1$, $n = 2$, or $n = 3$ —this control method targets the optimal operating point where ELMs are suppressed but confinement remains as strong as possible. Such techniques will be critical as future tokamaks run high-power, long-duration discharges and must keep ELM-induced heat pulses and particle fluxes within safe engineering limits. The demonstrations provided here indicate that a unified, adaptive amplitude approach can be generalized to diverse coil geometries and discharge scenarios, offering a strong foundation for advanced ELM control in next-generation devices.

It should be noted that full ELM suppression cannot always be achieved by simply increasing the amplitude of any applied RMP field. Instead, access to this regime also requires an adequate choice of RMP phasing, which is affected by various parameters, such as toroidal mode number, edge safety factor (q_{95}), and low pedestal collisionality [40, 41]. Phasing control proves valuable in this respect, as will be discussed in detail in section 3.

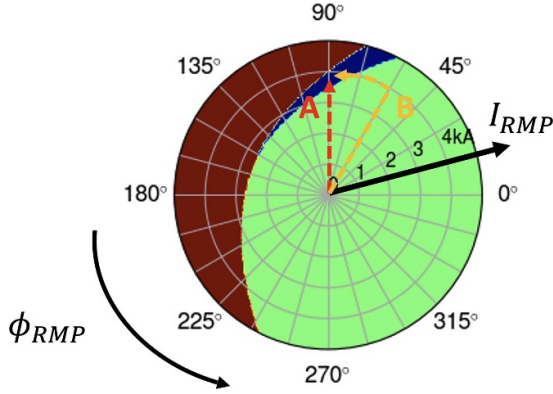


Figure 5. Illustrative polar plot output generated using IPEC, showing the value of phasing feedback control. The radial coordinate represents an RMP amplitude, assuming any other amplitudes are equal or a constant ratio (on KSTAR, this would imply $I_{UP} = I_{BOT}$ while I_{MID} is tied to them by being computed as a fixed ratio, and on DIII-D $I_{UP} = I_{BOT}$). The angular coordinate represents the RMP phasing, assuming all phasings are equal (on KSTAR this would imply $\phi_{TM} = \phi_{MB}$). Note that DIII-D only has a single phasing). Green areas in phase space represent regimes where the plasma remains ELMy. Red represents the region consistent with locked modes, and blue represents the region in phase space conducive to ELM suppression.

3. Expanding the ELM suppression window in feedback

There is no universally accepted physics explanation for the cause of the pedestal bifurcation into an ELM-suppressed state yet. However, a leading contender is the penetration of the applied RMP at edge resonant surfaces [40]. Using this idea, physics codes such as IPEC [42, 43] and GPEC [44] have been developed, which use information from the plasma to determine the domain in phase space of the RMP coils that are consistent with ELM suppression, with no suppression, and with locked modes. When limiting the degrees of freedom to a single amplitude and phasing, polar plots can be generated that show the ELM suppression window [45]. These predictions have been successfully compared against experimental observations. An example is presented in figure 5.

In practice, the maximum applicable RMP amplitude is limited, and in general, a low RMP amplitude is desired because of its minimal impact on performance. However, as shown in figure 5, the ELM suppression window can be very narrow at the lowest acceptable RMP amplitude, while it broadens for some greater amplitudes. More importantly, at a given constant RMP amplitude, changing the RMP phasing through the appropriate path in phase space can provide safer access to ELM suppression (indicated by orange curve B)—avoiding disruptions induced by locked modes. This means that controlling the phasing in feedback will expand the ability of the controller to achieve ELM suppression under a wider range of plasma parameters, it will enable the controller to enter ELM suppression more safely with margin in the ELM suppression window, and it will allow further plasma performance optimization in the ELM suppressed phase. Even though the specific (trigonometric) relationships used to compute the

total current requests for each power supply depend on the patch panel configuration, it is still helpful to provide a representative example to illustrate the concept.

The RMP current amplitude requests for the top, middle, and bottom rows of coils are computed by summing their feed-forward and feedback contributions:

$$I_{UP} = I_{UP,FF} + I_{UP,FB} \quad (1)$$

$$I_{MID} = I_{MID,FF} + I_{MID,FB} \quad (2)$$

$$I_{BOT} = I_{BOT,FF} + I_{BOT,FB}. \quad (3)$$

In order to be able to determine the phases of individual rows, first, the two total phasings are computed. Note that ‘total phasings’ refers to the relative phases between the associated rows of coils, where subscripts ‘T’, ‘M’ and ‘B’ refer to ‘Top’, ‘Middle’ and ‘Bottom’ rows respectively:

$$\phi_{TM} = \phi_{TM,FF} + \phi_{TM,FB} \quad (4)$$

$$\phi_{MB} = \phi_{MB,FF} + \phi_{MB,FB}. \quad (5)$$

The phases are then computed as follows:

$$\theta_{MID} = \theta_{MID,FF} \quad (6)$$

$$\theta_{UP} = \theta_{MID} - \phi_{TM} \quad (7)$$

$$\theta_{BOT} = \theta_{MID} + \phi_{MB}. \quad (8)$$

The total currents are then computed by multiplying the amplitude by a simple trigonometric function of the phase. As an illustrative example:

$$I_{i,j} = I_i \cos(\theta_i + \theta_j) \quad (9)$$

where:

- $i \in \{\text{TOP, MID, BOT}\}$ corresponds to the current amplitude I_i associated with each row of coils,
- $j \in \{0, 1, \dots, n_{\text{coil}} - 1\}$ indexes the n_{coil} toroidally distributed field coils in each row,
- θ_i is the toroidal phase offset for row i , and
- θ_j is the toroidal angle of the j th coil in the row.

Full 3D spectrum control where phasing feedback control is informed by real-time plasma measurements processed by a ML surrogate model of more complex offline physics-based models has shown promising results [37]. Full 3D spectrum control refers to the ability to control both the RMP amplitudes and phases simultaneously. A more rudimentary phasing control to explore its effect on ELM suppression shows valuable insights into how the time trace history can affect the ELM suppression window.

The phasing feedback control mechanism used for initial exploration is comprised of the same finite state machine as is used for amplitude feedback control. For an in-depth outline of the finite state machine design, the reader is referred to our previous publication [33].

First results from phasing feedback control using $n = 1$ RMP are presented in figure 6, which shows the time trace

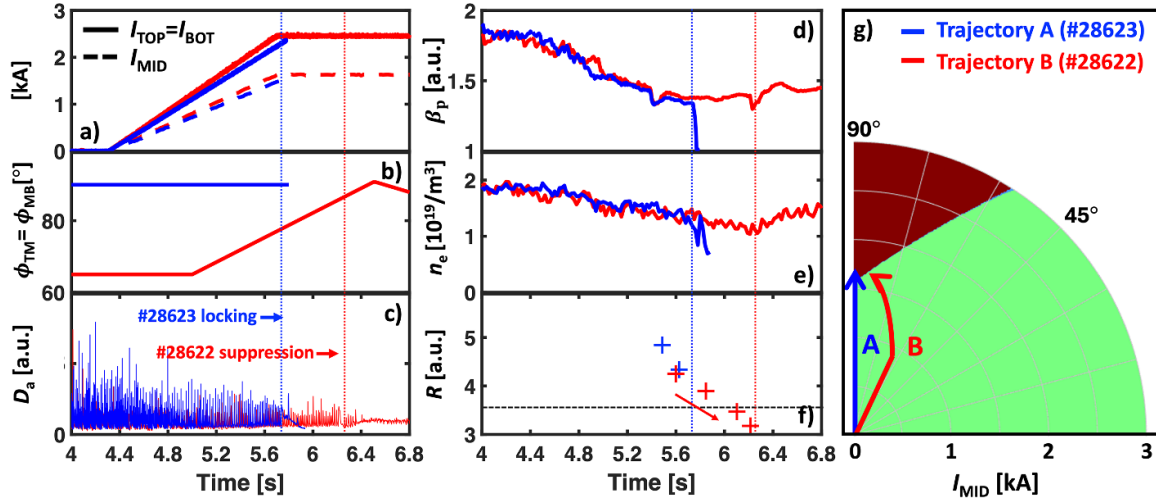


Figure 6. Time traces of (a) top (I_{TOP}), middle (I_{MID}), bottom (I_{BOT}) RMP coil currents, (b) RMP coil phasing between top-middle (ϕ_{TM}), middle-bottom (ϕ_{MB}), (c) D_α signal, (d) poloidal beta, (e) electron pedestal density, (f) ratio between core, edge resonant field (R), and (g) RMP establishment on the polar contour. The blue and red lines correspond to KSTAR discharge #28623 (case (A)) and #28622 (case (B)), respectively.

of two discharges (#28622 and #28623) whose plasma conditions are identical before the RMP application. Major plasma parameters are $I_p \sim 0.5$ MA, $B_T \sim 1.8$ T, $q_{95} \sim 5$ and $\kappa \sim 1.7$. In these discharges, ELM suppression is attempted by ramping up the RMP current with fixed 90 degrees phasing (#28623, A) and by modifying both amplitude and phasing (#28622, B) simultaneously. The trace of the RMP ramp for these cases is shown in figure 6(g). Here, the comparison illustrates that RMP establishment through trajectory B can safely access the ELM-free state (at 6.2 s) while trajectory A fails due to core-mode locking (at 5.65 s). Interestingly, RMP amplitude and phasing at ELM suppression access or locking in both cases are similar but have a different outcome, showing the benefits of ramping amplitude and phasing simultaneously. This is possible because the ratio (R) of core and edge response differs. The R value in this study is derived from the plasma response calculation, where the edge response is the resonant field component at the 5/1 surface (near the pedestal top) [41, 46] and the core component is the averaged resonant field over the $n = 1$ rational surfaces in the core region ($\psi_N < 0.8$) for a given 3D field configuration.

Because safe ELM suppression access requires a weak core and strong edge response, a small R is needed here. For the reference case, posterior analysis reveals that $R < 3.5$ is required for entering ELM suppression. Here, $R = 4.3$ in #28623 at 5.6 s exceeding the threshold, and therefore, a disruption occurs at 5.65 s when the RMP amplitude is large enough to drive core-locking. Phasing feedback control discharge #28622 has $R = 4.15$ at 5.6 s, which is slightly smaller than #28623 but still too large for suppression. Interestingly, R decreases to 3.0 before 6.2 s, falling below the threshold required for ELM suppression and thereby allowing access to the suppressed regime. As shown in figure 6(f), this decrease occurs gradually before the onset of suppression. The resulting lower R expands the ELM suppression window, as illustrated in figures 7(a) and (b), where suppression begins as early as 6.2 s. This favorable

change in R arises from a combination of reduced pedestal height, which degrades global beta and weakens the core resonant response due to diminished beta amplification, and lowered edge current density, which shifts the rational surface (near the pedestal top) radially outward, where the applied perturbation is stronger, resulting in an enhanced edge resonant field.

Figures 7(c) and (d) compares the edge pressure and current profiles for those time slices. Here, it can be seen that the pressure pedestal is smaller in #28622. This is because the plasma has more time to fully respond to the applied edge field, which degrades the pedestal. When the RMP amplitude and phasing are ramped up simultaneously from a low initial value, the core and edge field perturbations can increase more gradually compared to cases where the phasing is constant at 90 degrees. This is because the field perturbation increases with phasing, and thus, even if the RMP amplitude ramp rate is similar, as in discharges #28623 and #28622, the simultaneous slow increase of phasing in #28622 results in a more gradual rise in the overall field perturbation. This slower buildup allows the pedestal to respond to the applied field over a longer period before the core field perturbation reaches the locking threshold, enabling greater pedestal degradation.

As both edge current density (by reduced bootstrap current) decreases and pedestal degrades, it leads to a decrease in R . Therefore, the smaller R before accessing ELM suppression in #28622 can be considered an outcome of favorable pedestal profile evolution during the RMP ramp. Furthermore, increased pedestal degradation results in higher pedestal resistivity, facilitating the penetration of the perturbed field [47], easing the ELM suppression access. This analysis highlights the advantage of a slow RMP ramp-up, as it allows the pedestal to fully respond to the applied edge field and a gradual reduction in R with pedestal degradation. From this perspective, simultaneously ramping the RMP amplitude and phasing can be particularly useful for ITER operations. First, this approach

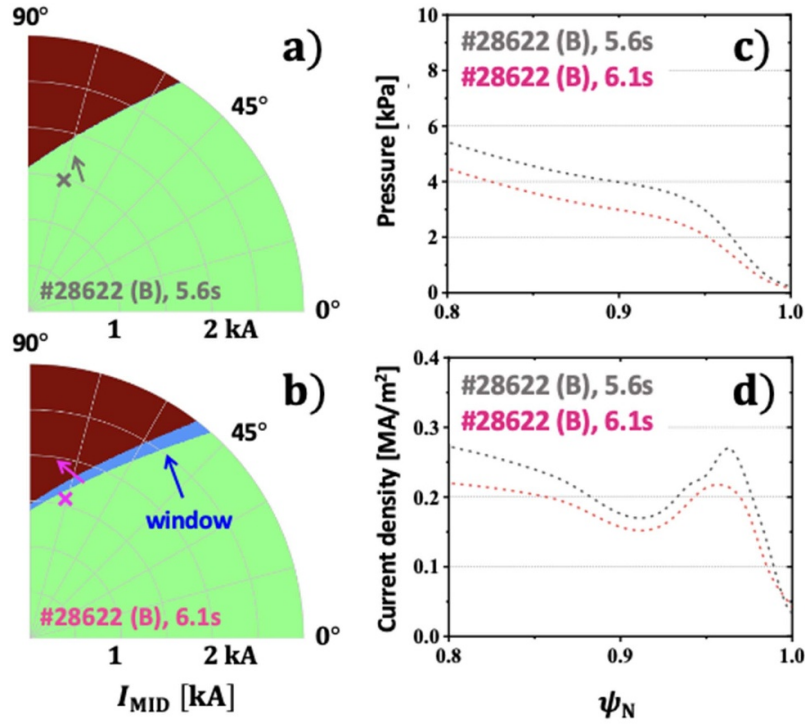


Figure 7. The polar contour plot of KSTAR discharge #28622 for (a) $t = 5.6$ s and (b) 6.1 s. The comparison of radial profiles of (c) total pressure and (d) current density.

results in a relatively slow increase in the effective field perturbation, even if the amplitude is ramped quickly. In addition, starting from a low phasing mitigates the risk of excessive core perturbation early in the ramp. As phasing gradually increases, it also becomes possible to naturally identify the optimal phasing for effective ELM control through feedback control, enabling flexible RMP optimization. Therefore, a simultaneous phasing and amplitude feedback control will be promising for advanced RMP control in a future device.

4. Toward complete ELM suppression

The feedback adaptive RMP ELM controller has proven successful in suppressing ELMs across different scenarios, yet it initially required at least one ELM to occur before optimizing the applied RMP amplitude [33]. This limitation arises because no real-time models exist to predict the minimum RMP level needed to maintain ELM suppression. While the single-ELM approach is tolerable in present devices, next-generation tokamaks like ITER may not afford even a single large ELM crash, as it could lead to significant damage and reduced device lifetime [5, 7].

To address this challenge, we improved the controller to aim for completely ELM-free discharges by suppressing the initial H-mode ELMs and anticipating losses of ELM suppression. Here we outline three key developments toward that goal: (i) early ELM suppression triggered by a ML-based LH-transition detector, (ii) robust precursor detection

and immediate ‘jump-scheme’ actuation, and (iii) an active ‘probing-scheme’ to dynamically adjust the RMP amplitude without incurring ELMs. These upgrades were tested on KSTAR and DIII-D to demonstrate improved ELM avoidance, enhanced performance, and pathways for future optimization.

4.1. Early ELM suppression enhanced by machine learning

A major drawback of previous approaches was that the RMP ramp-up began only after the plasma entered ELMy H-mode, exposing the device to initial ELMs. However, if RMP is applied too early, it can hinder the LH transition [48]. To mitigate this, we integrated a real-time ML-based LH transition detector [49] with the feedback adaptive RMP ELM controller. Once the detector identifies the LH transition, the controller immediately applies a rapid feed-forward RMP ramp to suppress emerging ELMs.

Figure 8 illustrates a typical discharge on KSTAR where the detector triggers the controller at the moment of transition. The controller ramps the RMP amplitude rapidly, effectively quelling most ELMs that normally appear just after the LH transition. Afterwards, the controller smoothly switches to feedback-driven RMP amplitude regulation. While a few sporadic ELM spikes may remain, confinement degradation is minimized, achieving record β_N values of approximately 2.69 at later times. An in depth discussion of this achievement is detailed in [50]. Further refinement of ramp rates and steady-state amplitudes could fully eliminate these residual ELMs.

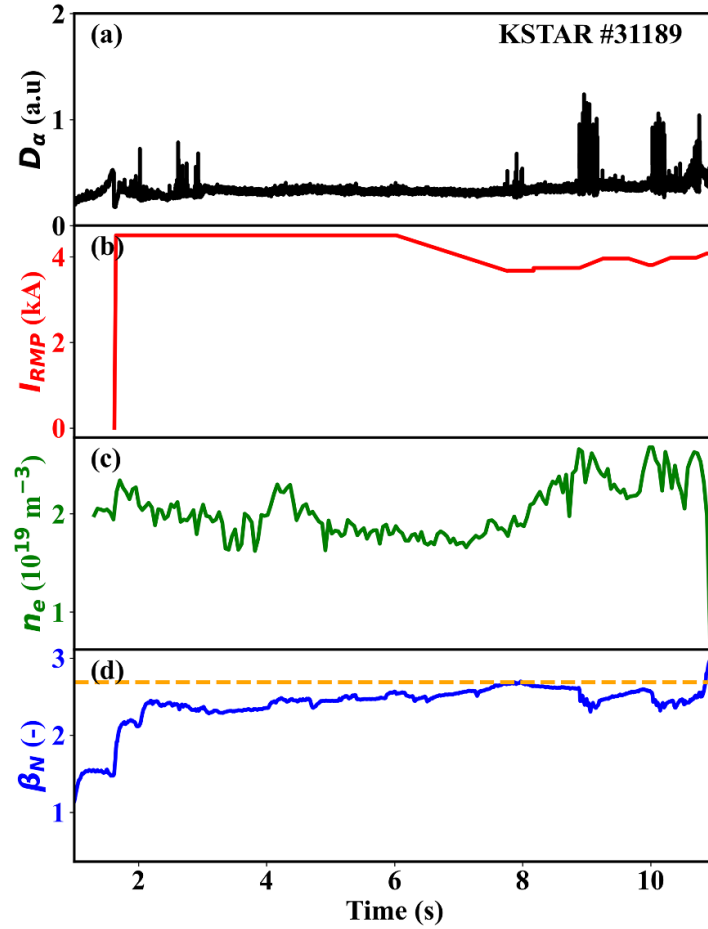


Figure 8. Experimental demonstration of early RMP ramp as feedback adaptive RMP ELM controller is triggered by ML-based LH-transition detector, followed by feedback optimization. The D_α trace is shown in panel (a). Panel (b) shows the requested RMP amplitude, panel (c) indicates the pedestal electron density and the bottom pane shows β_N in blue with the horizontal orange dashed line indicating $\beta_N = 2.69$. This scheme is successful at mitigating and avoiding most ELMs in the discharge. Optimized ramp rates and amplitudes could further improve these results.

4.2. Enhancing ELM suppression through precursor detection techniques

Instead of relying on a fully predictive model for RMP thresholds, a viable approach is to detect early signs of impending ELMs and correct the RMP amplitude preemptively. Through analysis of D_α and magnetic signals, we identified a consistent precursor signature—namely, a dip in D_α baseline accompanied by a drop in the standard deviation of Mirnov signals—occurring 10–20 ms prior to an ELM crash. Such a lead time is sufficient for real-time feedback.

Although Mirnov probes offer clear precursor signatures, real-time—qualified Mirnov data was not always available at KSTAR; thus, our initial real-time detector relies only on D_α measurements. As summarized in figure 9 and table 1, the precursor detector exhibits high reliability, with an 88% true positive rate in predicting imminent ELM crash events across multiple KSTAR discharges.

To robustly differentiate between baseline plasma behavior, precursor onset, and full ELM crashes, we adopted an Interacting Multiple Model Kalman Filter (IMMKF) approach [35, 36]. It runs three parallel Kalman filters—each

modeling a different ELM-related plasma state—and continuously updates the likelihood of each model given real-time measurements (see figure 10). When the ‘Precursor’ mode probability exceeds a threshold, the detector flags an imminent ELM suppression loss. This design can be adapted to other devices by replacing sensor-specific models.

Figure 11 highlights the final decision logic that mitigates false positives. Precursor flags raised outside ELM-suppressed intervals are ignored, ensuring minimal unnecessary actuation. Tests on KSTAR show that 10–15 ms of lead time is typically achievable, sufficient to adjust RMP amplitude and maintain ELM suppression.

4.3. Suppressing imminent loss of ELM suppression through jump-scheme

When the RMP amplitude is minimized for optimal confinement, even small reductions below the suppression threshold can cause an ELM to reappear. As soon as the precursor detector signals imminent loss of ELM suppression, our new jump-scheme boosts the RMP amplitude quickly in a step-like

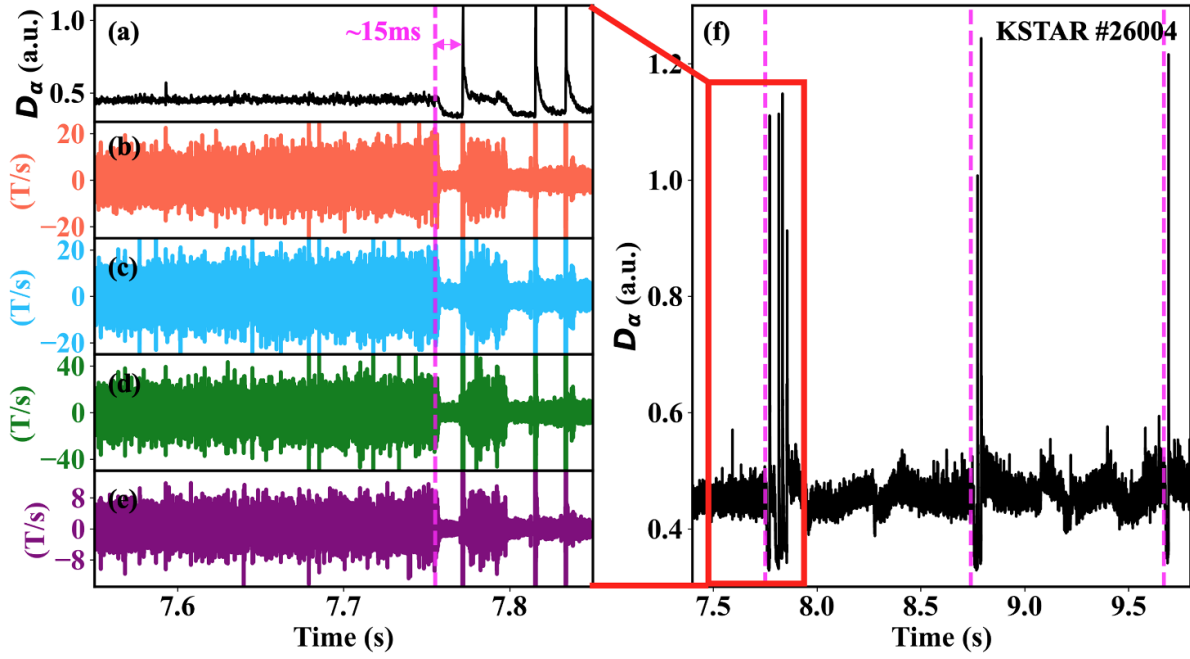


Figure 9. Examples of apparent precursors to loss of ELM suppression on KSTAR. The right-hand pane shows a D_α trace that exhibits three moments where ELM suppression is lost. Each time, a precursor can be identified, as labeled by the vertical dashed magenta lines. The panes on the left-hand side show the signal dynamics associated with the first loss of ELM suppression in greater detail. Panes (b)–(e) show data from Mirnov probes MC1P10, MC1P12, MC1P13 and MC1P15 respectively.

Table 1. Precursor detector performance assessment conducted on 10 KSTAR discharges taken from the 2020 and 2022 campaigns. Even though not reported in the table, the vast majority of the time slices where no precursor to loss of ELM suppression takes place are correctly ‘ignored’ by the detector. The performance results in the table represent the true positives (TP), false positives (FP), and false negatives (FN).

	Suppression loss events	TP	FP	FN
No.	34	30	0	4
Percentage	100%	88.24%	0%	11.76%

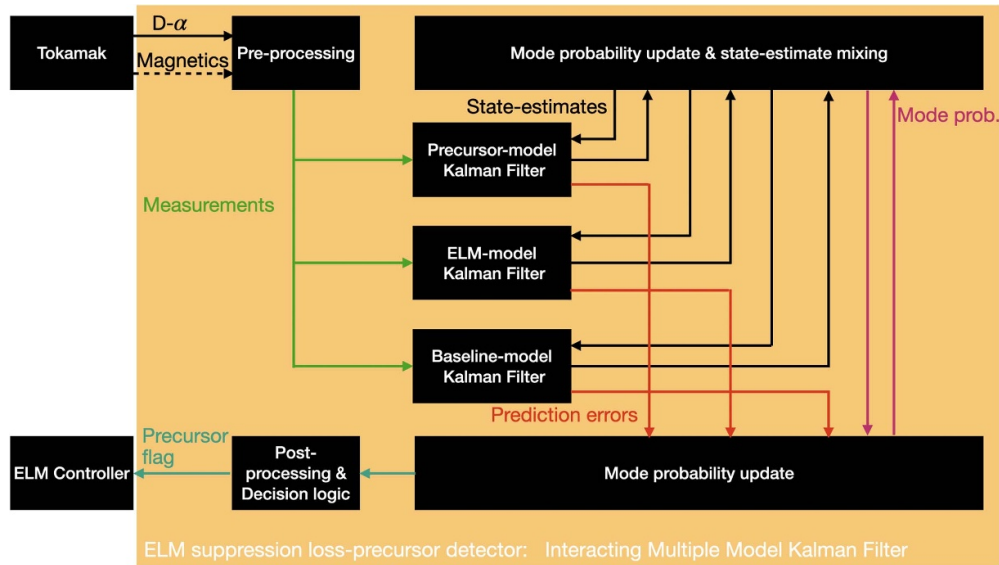


Figure 10. Simplified schematic representation of IMMKF used for detection of precursors to loss of ELM suppression. After preprocessing the measurements, they are fed into three Kalman filters that are executed in parallel. Each of these Kalman filters uses its own model (Baseline, ELM, Precursor) to predict expected signal features. The predictions are compared against the measurements, and this information is used to compute the model likelihood for each of the models. The results are provided to the feedback adaptive RMP ELM controller so that action may be taken.

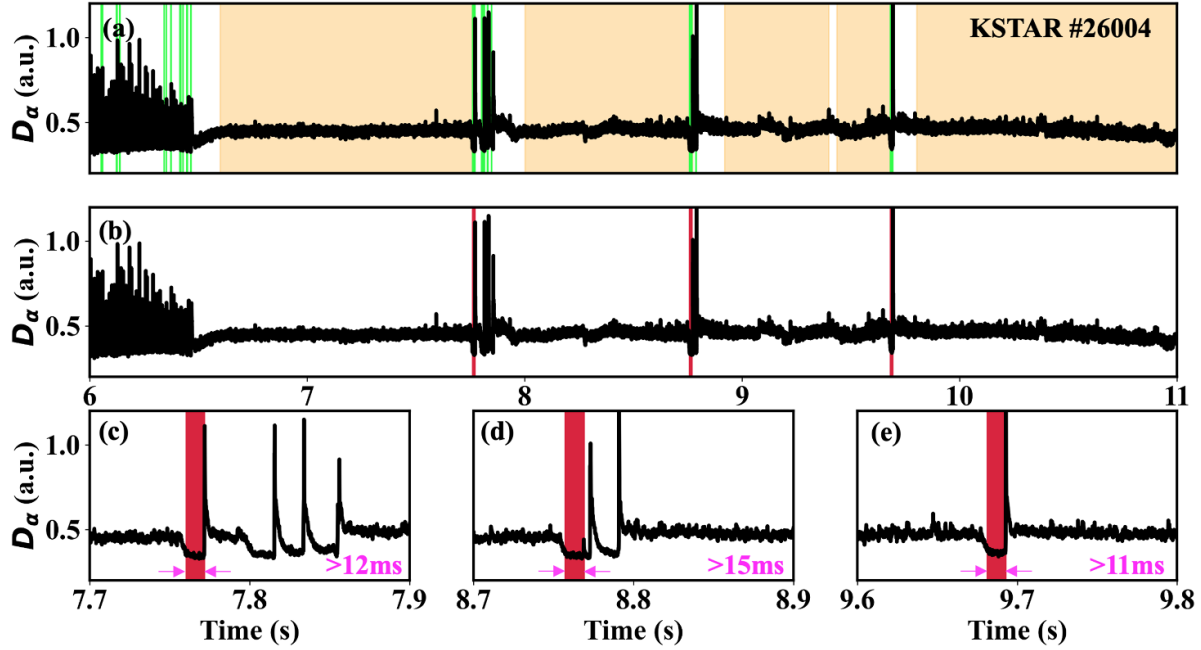


Figure 11. Final precursor flag determination after decision logic on KSTAR #26004. This figure shows an example of precursor detector performance on KSTAR. Panel (a) shows D_α in black, raw precursor detections in green, and ELM suppression as detected by the controller in shaded orange. In panel (b), the actual precursors as determined by the controller are shown in red. This subset of actual precursors is pruned of false positives by requiring ELM suppression during the detection of any precursor.

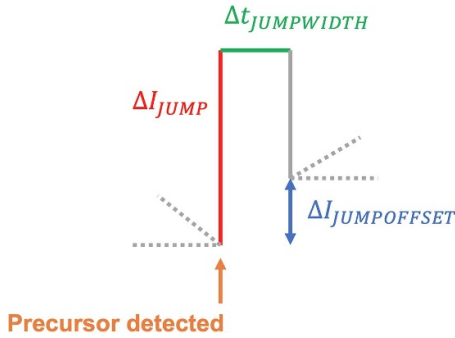


Figure 12. Cartoon representation of dynamics of feedback-controlled actuator signal when the controller applies a jump. When a precursor is detected, as indicated by the orange arrow, the requested RMP is increased in a step-like manner by an amount that is a predetermined user input in PCS, $\Delta I_{JUMP}(t)$. Next, the jump is sustained for $\Delta t_{JUMPWIDTH}(t)$ and then reduced to its original level plus a potential offset $\Delta I_{JUMPOFFSET}(t)$. After the jump has been completed, the controller continues its regular behavior.

fashion (figure 12). The jump is maintained for a configurable interval before returning to a slightly higher baseline than before.

Figure 13 demonstrates how the jump-scheme was tested on KSTAR discharges to suppress upcoming ELMs triggered by inadequate RMP amplitude. In some cases, the ELM was fully avoided, while in others, the amplitude jump was insufficient or occurred too late. By scanning jump amplitude, duration, and offset (figure 14), an operational space emerges where ELMs can be prevented with minimal plasma

performance penalties. The optimal jump shape likely depends on machine-specific factors such as available coil configurations and target plasma parameters.

4.4. Avoiding ELM suppression loss through probing-scheme

Long-pulse discharges introduce additional complexity: a lower bound for the RMP amplitude is often raised whenever ELM suppression is lost. If occasional, non-RMP-related ELMs occur, the adaptive lower bound can become overestimated, locking the system at unnecessarily high RMP amplitude and limiting further optimization.

We therefore introduced an active probing scheme to safely ‘test’ whether the RMP amplitude can be reduced without losing suppression. As depicted in figure 15, the controller briefly lowers the RMP by a set amount, observes whether a precursor or ELM arises, and then restores the amplitude to its original level (possibly offset). If no ELM or precursor is detected, the controller lowers the adaptive bound accordingly.

Figure 16 shows initial KSTAR demonstrations where a larger probe occasionally triggered ELMs, while a smaller probe size in figure 17 avoids those effectively. These results underline the importance of systematically exploring probe size, width, offset, and timing. In some cases, even short downward pulses can reveal new plasma conditions that permit lower RMP without ELMs, which is especially valuable for optimizing confinement in long-pulse discharges.

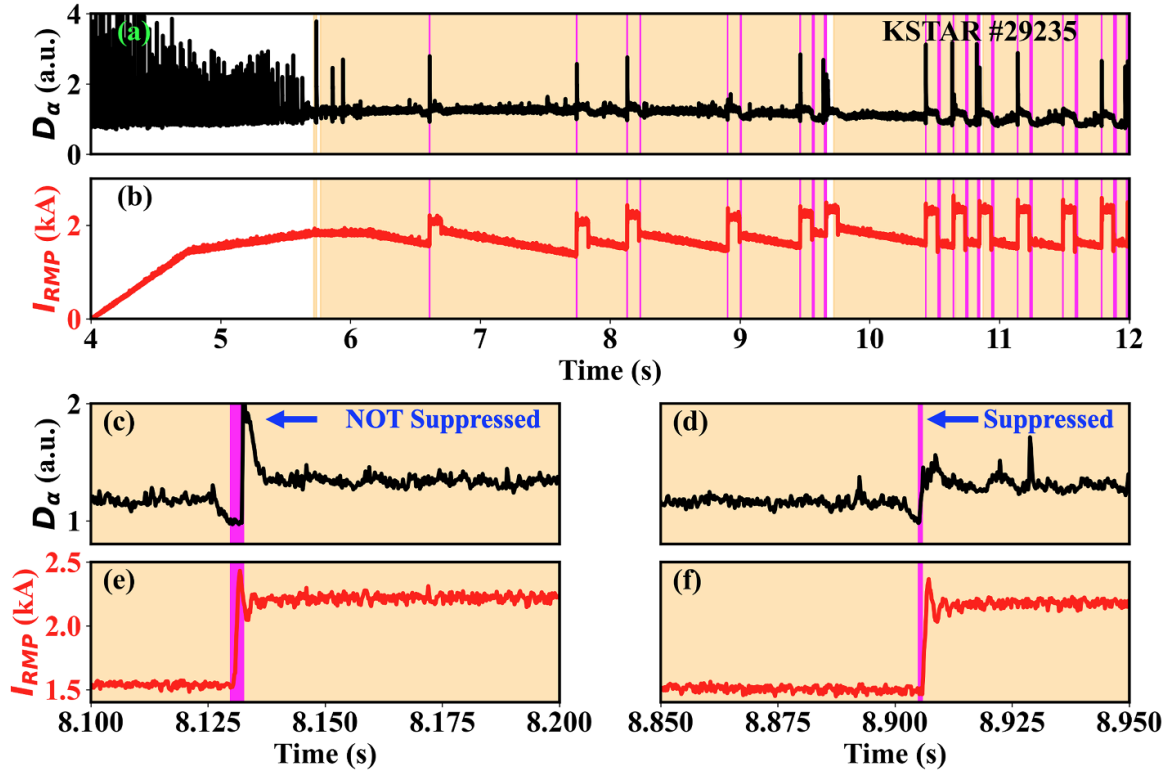


Figure 13. Example discharge used in the scan to optimize the jump waveform parameters on KSTAR. The D_α trace is shown in figure 13(a), alongside ELM suppression and precursors as detected by the controller in shaded orange and magenta, respectively. The RMP amplitude that exhibits the jumps is shown in panel (b). Panel (c), (d) and (e), (f) show the same signals as panel (a) and (b) respectively, but zoomed in to compare a case of successful ELM suppression through application of a jump versus a failed attempt earlier in time.

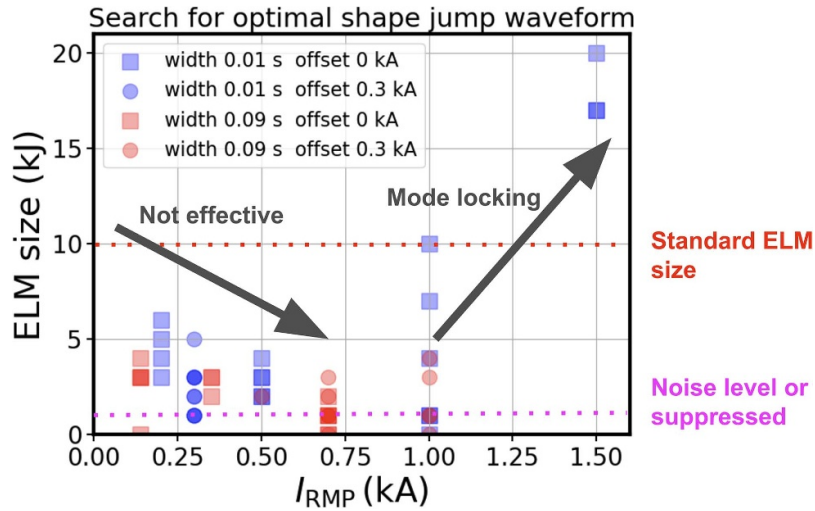


Figure 14. The width, height, and offset of the jump waveform have been varied to gain insight into any trends toward an optimal shape. In this preliminary overview, data from 5 KSTAR discharges are included. The color of the markers reflects the jump width, while the marker symbol itself represents the offset. The vertical axis depicts the ELM size, while the horizontal axis represents the jump height. There seems to be a trend indicating an optimal jump height between 0.75 and 1 kA.

5. Integrating control strategies toward long-pulse ELM suppression

To extend ELM-free operation over many seconds, the above control components—early activation, precursor-based jump-scheme, and active probing—must be coordinated. Figure 18

compares two long-pulse KSTAR discharges. In the reference shot, the adaptive lower bound grew so high that the RMP current saturated, restricting further optimization. In contrast, applying a probing scheme in the companion shot enabled a partial reduction of the overly conservative RMP amplitude, improving confinement (as evidenced by higher β_p) while

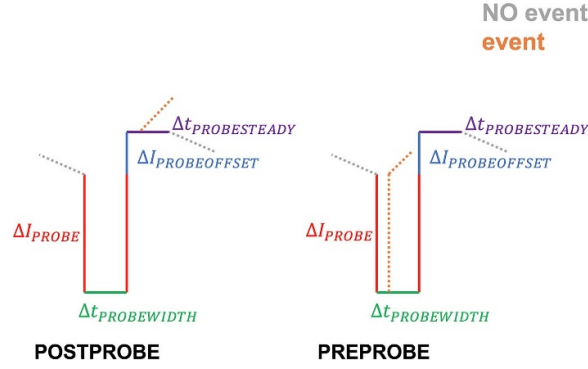


Figure 15. Cartoon representation of dynamics of feedback-controlled actuator signal when the controller applies a probe. A full probe starts with a sudden reduction in the requested RMP amplitude, amounting to user input $\Delta I_{\text{PROBE}}(t)$. Then, similar to the jump-scheme, the RMP request is held steady at this reduced amplitude for $\Delta t_{\text{PROBEWIDTH}}(t)$ time, before the requested amplitude reverts back to its original level plus a potential offset. Then, another timer is initiated. After $\Delta t_{\text{PROBEWAIT}}(t)$ has elapsed and no events of interest have been detected, the probe was successful, and the adaptive lower bound may be lowered. When POSTPROBE method is used, the controller ignores any events that may happen during the downward pulse, whereas PREPROBE method would monitor events during the downward pulse, immediately undoing it in case a precursor is detected. This intended behavior is illustrated by the orange dashed curves.

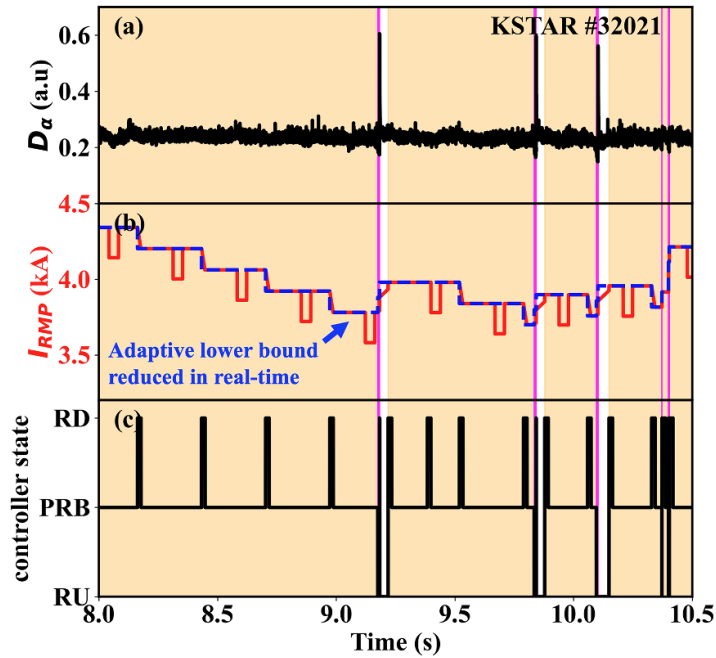


Figure 16. Experimental demonstration of active probing scheme in KSTAR: initial active probing tests to reduce lower bound. Panel (a) shows the D_α -trace in black, and the shaded orange and magenta areas represent ELM suppression and precursors as detected by the controller. The controller state is depicted in panel (c) and the resulting feedback signal is shown in panel (b).

successfully avoiding large ELMs. Only a few sporadic small, benign ELMs—well below typical ELM thresholds—were observed, consistent with the goal of minimizing heat loads to PFCs without degrading plasma performance.

A key challenge in applying RMPs for ELM suppression is the potential for non-uniform heat flux distribution on plasma-facing components, particularly at the divertor targets. By adjusting the phase relationships among different RMP coil rows, it is possible to rotate or modulate the resulting magnetic perturbation field. This phasing approach effectively ‘sweeps’ the footprint of the heat flux across the divertor surface over time, preventing localized overheating and thereby mitigating erosion or other damage.

Preliminary KSTAR experiments have demonstrated that a rotating RMP phasing, superimposed on amplitude feedback, can be employed, which should help reduce asymmetric heat loads (figure 19). By continuously changing the relative coil phasing, the resulting lobes in the magnetic field topology move around the torus, dispersing the net energy flux. Although this rotation adds operational complexity, the benefit is a more uniform divertor temperature profile during ELM-suppressed phases.

Looking ahead, combining phase control with the ‘jump-scheme’ and ‘probing-scheme’ outlined in the previous sections could further optimize long-pulse operation. While amplitude feedback alone ensures ELM suppression, carefully

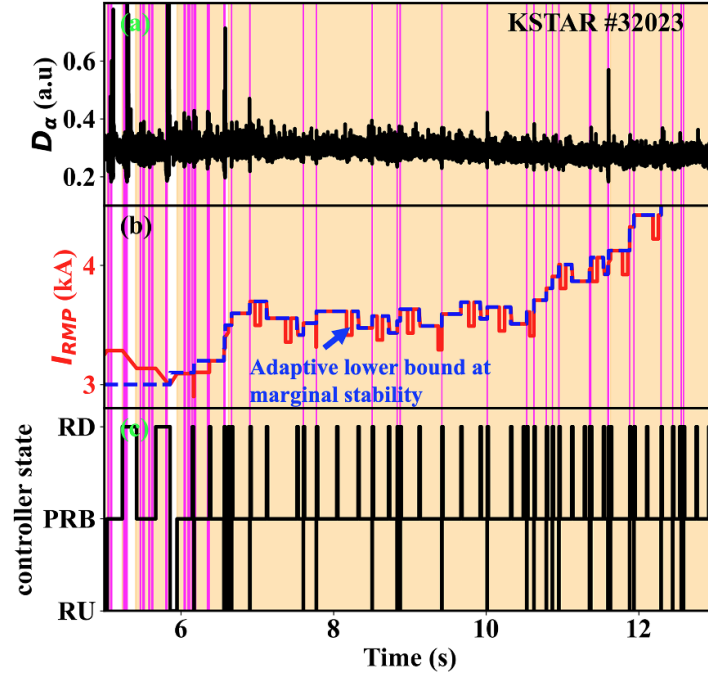


Figure 17. Experimental demonstration of active probing scheme in KSTAR: initial active probing tests to reduce lower bound. Panel (a) shows the D_α -trace in black, and the shaded orange and magenta areas represent ELM suppression and precursors as detected by the controller. The controller state is depicted in panel (c) and the resulting feedback signal is shown in panel (b). With these different probing waveform settings, the active probing-scheme is effective at avoiding ELMs at the marginal stability level. However as a result, the adaptive lower bound (indicated by the blue dashed curve) is not lowered as much as in figure 16.

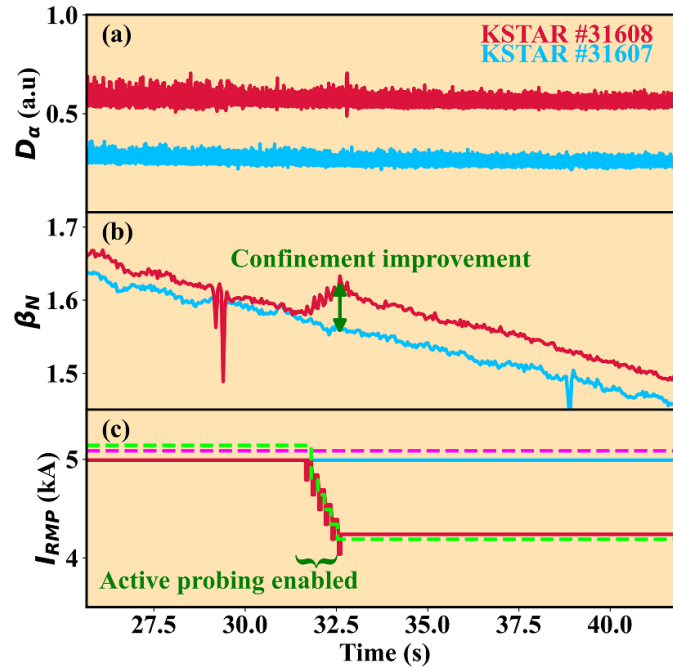


Figure 18. Experimental demonstration improved long pulse ELM suppression performance through application of the active probing scheme. Panel (a) shows the D_α trace for the reference discharge (#31607) in blue and the probing discharge in red (#31608). Note that the signal for the probing shot is offset with respect to the reference discharge for visibility purposes. In panel (c), relevant RMP amplitude waveforms are shown for both discharges. In the reference discharge, the adaptive lower bound (dashed magenta) eventually exceeds the maximum current request, forcing the RMP amplitude (blue) to remain at its maximum allowed value. The active probing scheme is active for a second after more than 30 s into the discharge. During this period, several probes are executed (red) and the adaptive lower bound is lowered accordingly (dashed green). In panel (b), β_N is observed to deviate from the reference starting when the probing scheme is activated. As the RMP is reduced during the probing period without causing ELMs, confinement is seen to increase as intended.

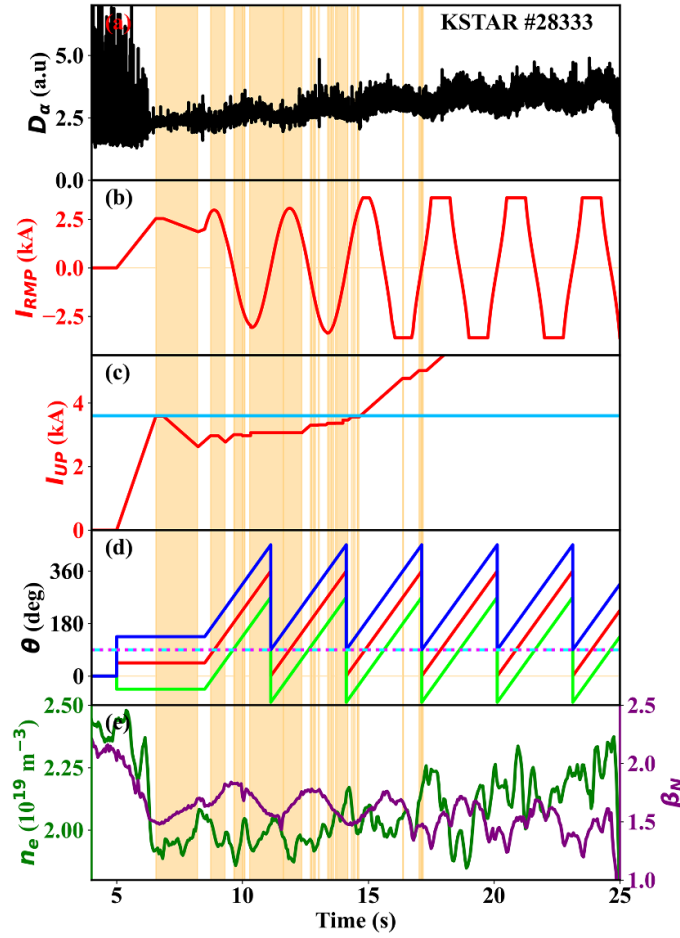


Figure 19. Experimental demonstration of feedback-controlled ELM suppression capability using rotating RMPs paving the way for toroidal heat load distribution. Panel (a) shows the D_α trace in black, and ELM suppression as detected by the controller in shaded orange. Panel (b) shows an example total RMP current request, which is calculated using equations of the form shown in equation (9), and the signals shown in panel (c) and (d): in panel (c) the raw feedback computed RMP amplitude is shown in red, and the maximum allowed current request in light blue. In panel (d) the feed forward phase of the middle row is shown in red. The derived phasings of the top and bottom row are depicted in blue and green respectively, computed using the constant phasings shown in magenta and cyan. The rotation can be observed as the sinusoidal behavior of the signal. Eventually the sinusoidal behavior seems to obtain flat extrema, which was caused by the current request exceeding the applied maximum current request. Panel (e) shows the pedestal electron density and β_N .

timed adjustments of RMP phasing may preserve plasma performance by minimizing negative impacts on core confinement and pedestal density. Ongoing investigations will focus on mapping how different phasing angles and coil configurations affect key metrics, such as pedestal pressure gradient and local divertor temperature rise.

Additionally, when combined with rotating phasing, the probing scheme can further reduce the adaptive RMP bound over time. In long pulses, plasma-facing components heat up and the discharge characteristics evolve, potentially lowering the amplitude threshold needed for stable ELM suppression. By detecting these changes through well-timed probing, the controller safely lowers the lower bound, maximizing confinement and reducing coil current demand over extended periods.

6. Remaining challenges in ELM suppression

The improvements described offer promising avenues toward sustained ELM-free operation in ITER-scale devices, but challenges remain. For instance, sawtooth-induced ELMs can appear as sharp spikes on D_α diagnostics, triggering the controller's ELM detection algorithm. This misclassification leads to an erroneous increase in RMP amplitude, under the assumption that a loss of edge stability occurred. In reality, these ELMs are triggered by core sawtooth crashes and are not suppressible by increased RMP, rendering the control action ineffective. Figure 20 illustrates this limitation: the D_α spike results from a sawtooth-induced ELM, which prompts an unnecessary controller response. Differentiating

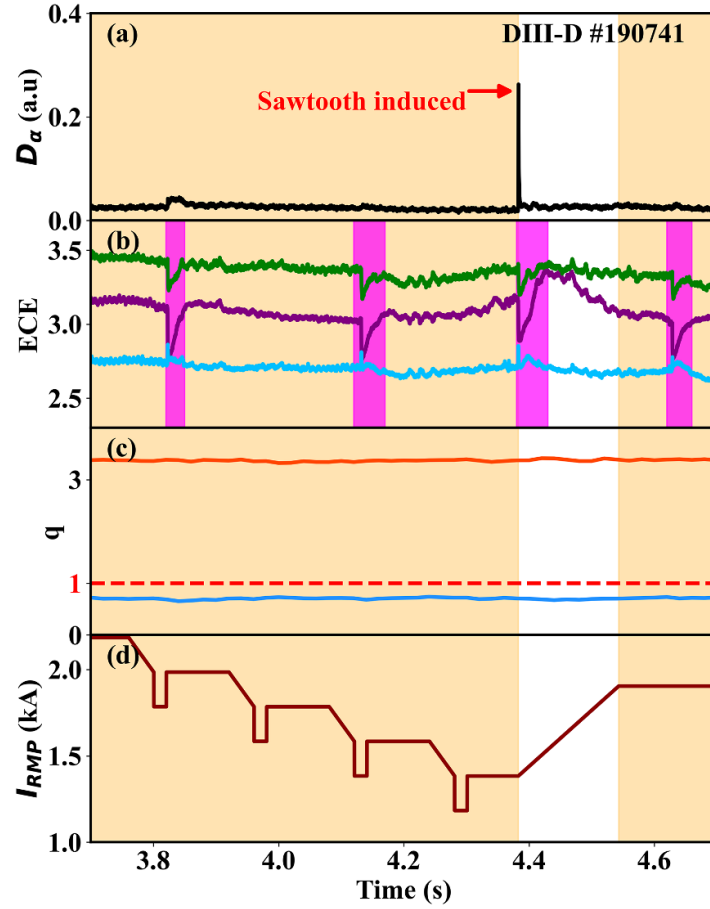


Figure 20. A sawtooth-induced ELM appears as a spike on D_α in panel (a), triggering the controller’s ELM detection and leading to an unnecessary increase in RMP amplitude. Panel (b) shows ECE channels confirming the sawtooth inversion radius, while panel (c) shows $q_{\min} < 1$. The bottom pane illustrates the resulting actuation by the controller.

edge-instability-driven ELMs from other MHD events in real-time requires more sophisticated detectors or multi-diagnostic approaches. In addition to sawteeth, other transient phenomena may also perturb the D_α signal in ways that lead the controller to incorrectly infer an edge-instability-driven ELM. Enhancing the robustness of the ELM detection algorithm—so it can reliably distinguish such events—will likely require a multi-diagnostic approach and represents important future work.

Additionally, the interplay between RMP geometry and transport physics must be fully characterized to ensure robust ELM suppression at minimal cost to confinement. Future work will involve large multi-parameter scans, machine-learning—accelerated model building, and further hardware integration to refine the synergy of jump-scheme, probing-scheme, and phasing control. Ultimately, achieving truly uninterrupted ELM-free H-mode discharges in the next generation of fusion devices will require a close interplay of diagnostics, advanced control algorithms, and real-time capable physics models.

7. Conclusion

In this paper, we have extended our previously published amplitude-based RMP ELM controller with multiple new strategies that broaden its operational window and enhance device flexibility. The demonstration of multi- n capability on both KSTAR ($n=2$) and DIII-D ($n=3$) confirms that the amplitude-feedback logic can be applied robustly across distinct toroidal mode numbers, in line with anticipated ITER scenarios. Introducing phasing control has further enlarged the ELM suppression window at moderate RMP amplitudes, providing a safer route to ELM-free conditions and helping to avoid locked-mode boundaries. Additionally, initial tests of rotating phasing have shown that sweeping the field perturbation toroidally can distribute divertor heat loads more uniformly, making it a viable tool to protect plasma-facing components during long-pulse operation. Beyond phasing, a jump-scheme was implemented to preemptively counter impending losses of ELM suppression by raising the RMP

amplitude when ELM precursors are detected. A complementary probing-scheme allows the controller to lower the RMP amplitude systematically and operate near the marginal stability boundary without inducing RMP-suppressible ELMs, thereby preserving optimal confinement as much as possible within the limitations of the current controller. Taken together, these new features—multi- n operation, phasing control, rotating fields, jump-scheme, and probing-scheme—enable more precise real-time regulation of the RMP fields over extended discharges. By integrating them into the existing controller design, we have confirmed that the same control algorithm can be ported to different devices with only minor interface modifications, underscoring the framework's adaptability.

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